Digital Image Processing Lab 06 Report (MATLAB)

Project Title:

Smart Image Enhancement & Analysis System

Course : Digital Image Processing Lab
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OBJECTIVE

The objective of this project is to design and implement a complete image enhancement system using MATLAB. The system improves low-quality images by applying multiple digital image processing techniques including transformations, sampling, quantization, and histogram processing.

Phase 1: Image Acquisition & Understanding

- Input image loaded using imread()
- Converted RGB image to grayscale using rgb2gray()
- Displayed image properties:
    - Resolution
    - Data type
    - Sample matrix values

Phase 2: Sampling & Quantization

- Image resized using scaling factors:
    - 0.25x, 0.5x, 1x, 1.5x, 2x
- Bit depth reduced:
    - 8-bit
    - 4-bit
    - 2-bit
- Observed loss of detail and pixelation

Phase 3: Geometric Transformations

- Applied:
    - Rotation (30°, 45°, 60°, 90°, 120°, 150°, 180°)
    - Translation
    - Shearing
- Attempted inverse transformation for restoration

Phase 4: Intensity Transformations

- Negative transformation

- Log transformation
- Gamma correction ($\gamma = 0.5, 1.5$)

Observation:

- Gamma < 1 → image becomes brighter
- Gamma > 1 → image becomes darker

Phase 5: Histogram Processing

- Histogram plotted for original image
- Histogram equalization applied
- Contrast improved after processing

Phase 6: Final System Pipeline

Input Image → Grayscale → Processing → Enhancement → Output Image

TASK :

CODE :

```
%% STEP 1: Load Image
img = imread('img interpol.JPG');
figure, imshow(img), title('Original Image');
%% STEP 2: Grayscale
gray = rgb2gray(img);
figure, imshow(gray), title('Grayscale Image');
disp('Resolution:');
disp(size(gray));
disp('Data Type:');
disp(class(gray));
%% STEP 3: Sampling (Resize)
half = imresize(gray, 0.5);
quarter = imresize(gray, 0.25);
double_img = imresize(gray, 2);
```

```
figure, imshow(half), title('0.5 Scale');
figure, imshow(quarter), title('0.25 Scale');
figure, imshow(double_img), title('2x Scale');
%% STEP 4: Quantization
img_8bit = gray;
img_4bit = floor(double(gray)/16)*16;
img_2bit = floor(double(gray)/64)*64;
figure, imshow(uint8(img_8bit)), title('8-bit Image');
figure, imshow(uint8(img_4bit)), title('4-bit Image');
figure, imshow(uint8(img_2bit)), title('2-bit Image');
%% STEP 5: Transformations
% Rotation
rot = imrotate(gray, 45);
figure, imshow(rot), title('Rotated 45°');
% Translation
tform = affine2d([1 0 0; 0 1 0; 50 30 1]);
trans = imwarp(gray, tform);
figure, imshow(trans), title('Translated Image');
% Shearing
shear = affine2d([1 0.5 0; 0 1 0; 0 0 1]);
sheared = imwarp(gray, shear);
figure, imshow(sheared), title('Sheared Image');
%% STEP 6: Intensity Transformations
% Negative
negative = imcomplement(gray);
figure, imshow(negative), title('Negative Image');
% Log Transformation
c = 255 / log(1 + max(double(gray(:))));
log_img = uint8(c * log(1 + double(gray)));
```

```
figure, imshow(log_img), title('Log Transformation');
% Gamma Correction
gamma05 = imadjust(gray, [], [], 0.5);
gamma15 = imadjust(gray, [], [], 1.5);
figure, imshow(gamma05), title('Gamma 0.5 (Bright)');
figure, imshow(gamma15), title('Gamma 1.5 (Dark)');
%% STEP 7: Histogram Processing
% Original Histogram
figure, imhist(gray), title('Original Histogram');
% Histogram Equalization
equalized = histeq(gray);
figure, imshow(equalized), title('Equalized Image');
figure, imhist(equalized), title('Equalized Histogram');
%% STEP 8: FINAL SYSTEM
final = histeq(imadjust(gray, [], [], 0.5));
figure, imshow(final), title('Final Enhanced Image');
```

OUTPUT:

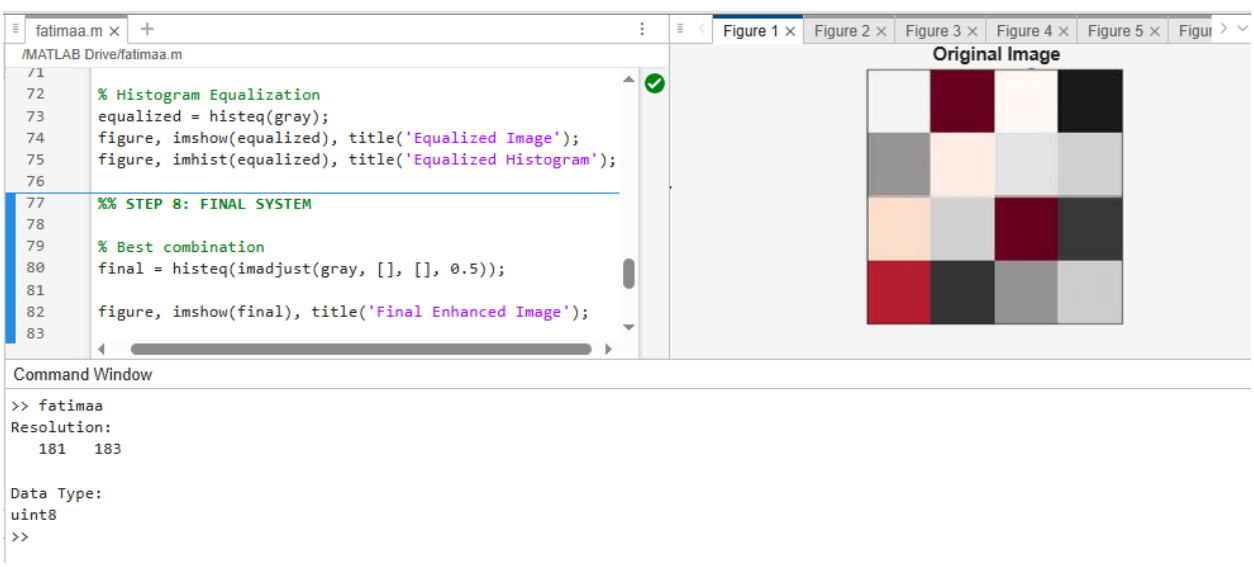


FIGURE 1:

FIGURE 2:

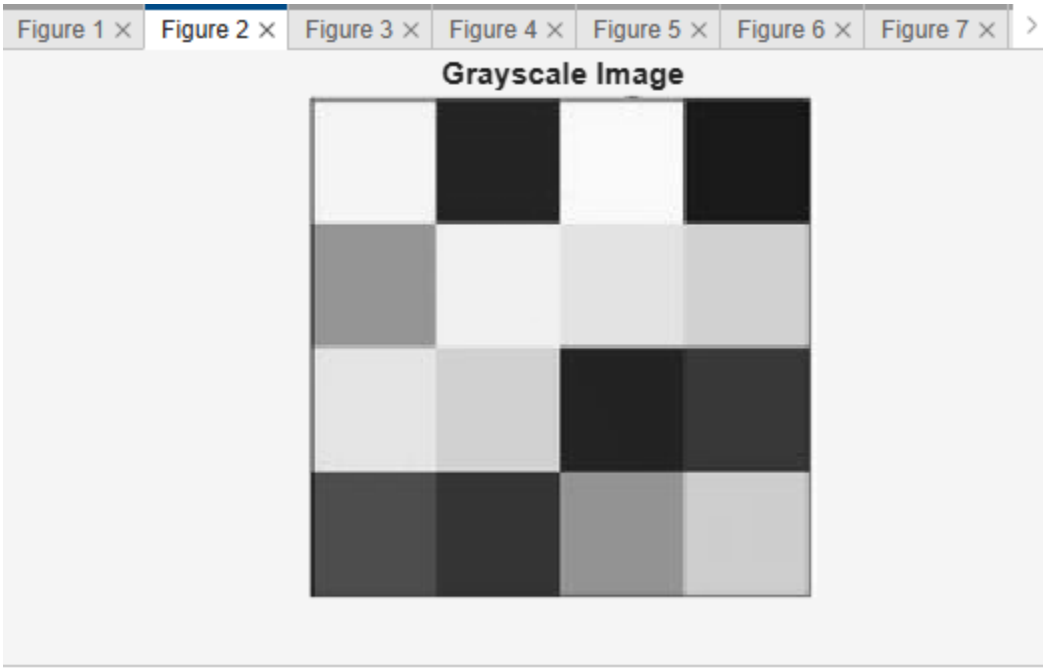


FIGURE 3:

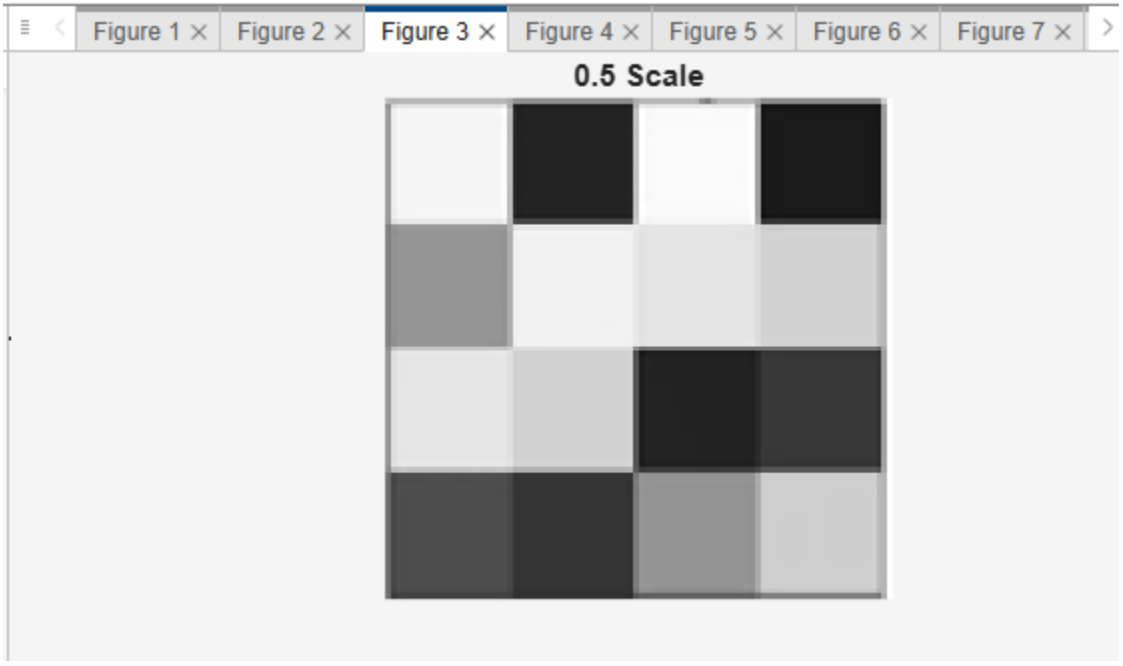


FIGURE 4:

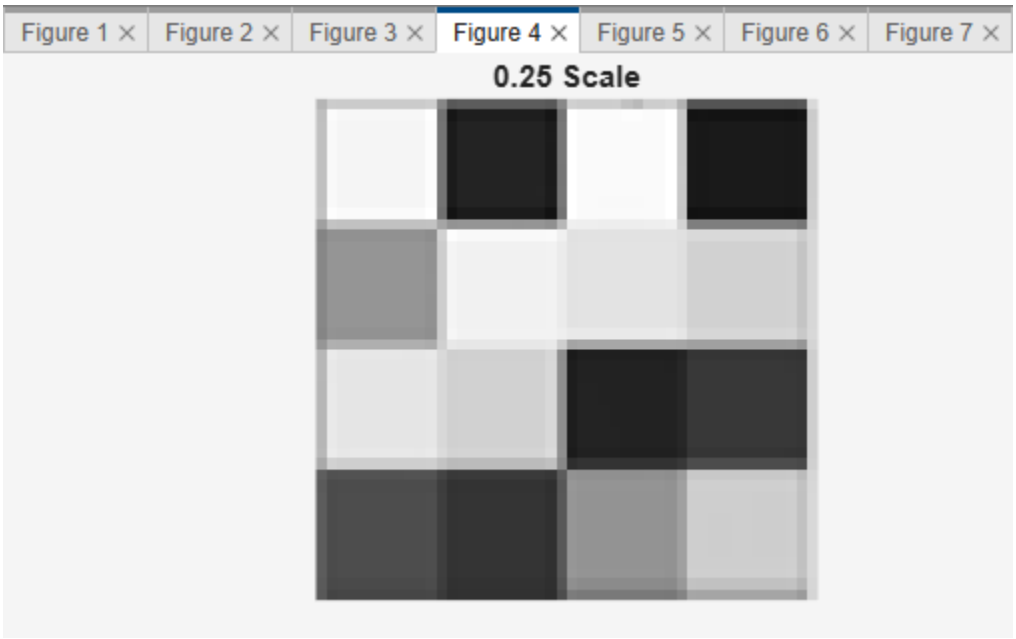


FIGURE 5:

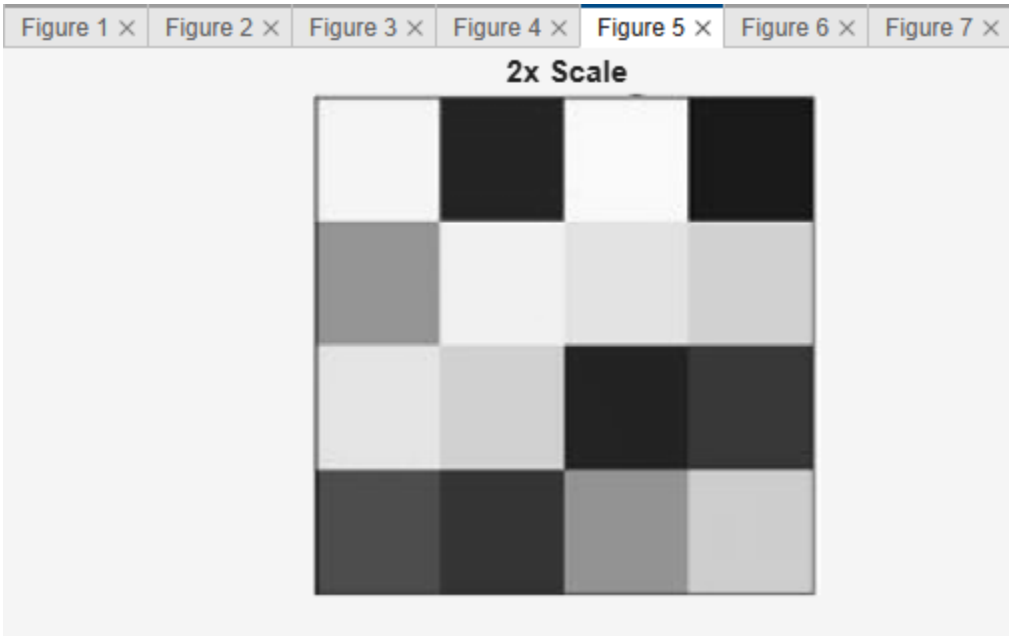


FIGURE 6:

FIGURE 7:

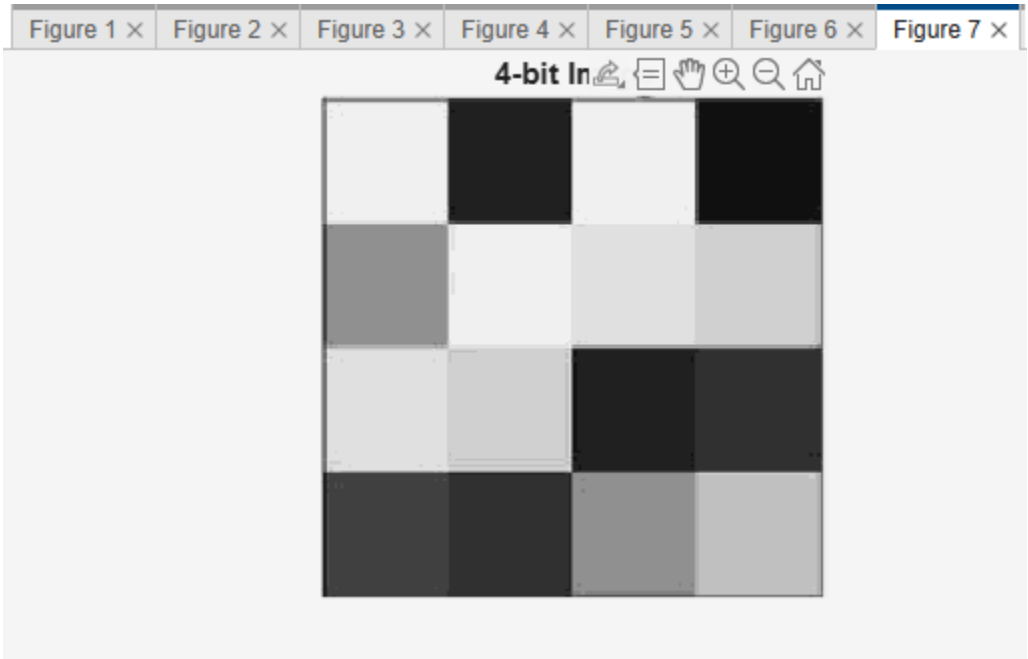


FIGURE 8:

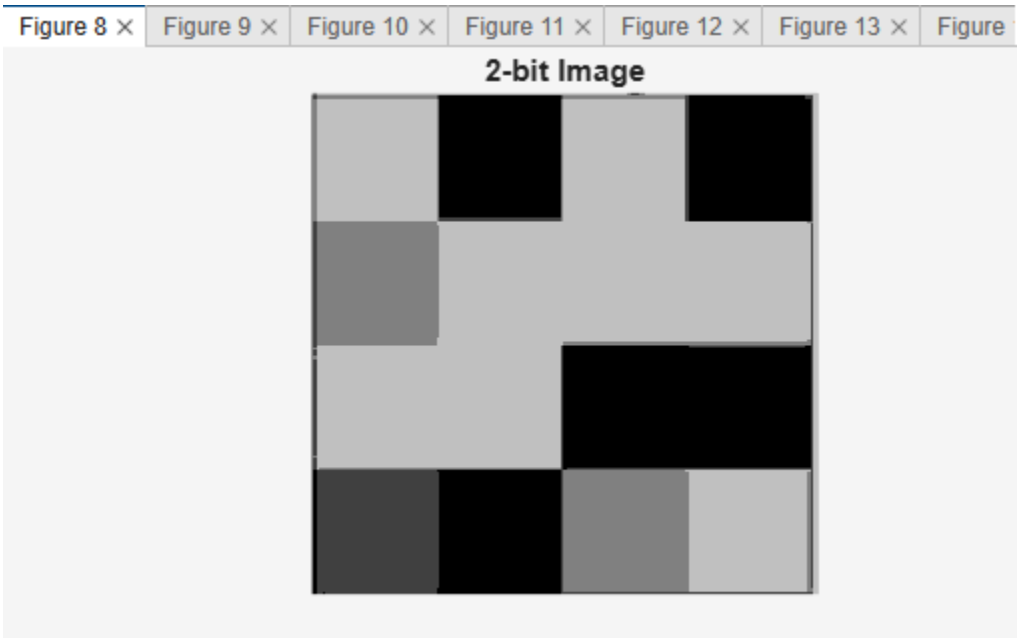


FIGURE 9:

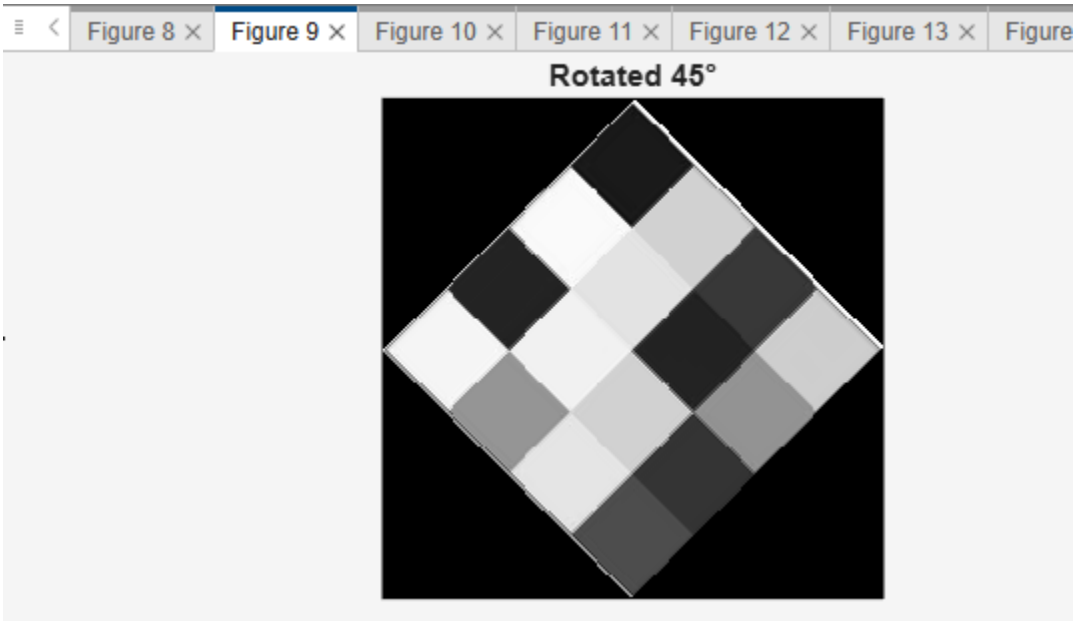


FIGURE 10:

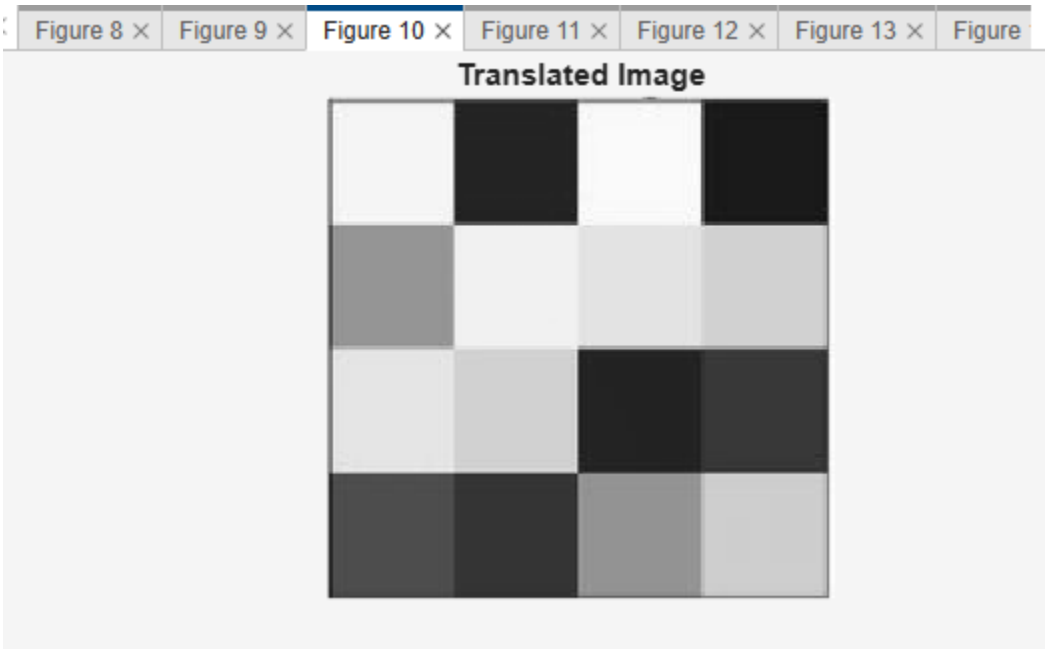


FIGURE 11:

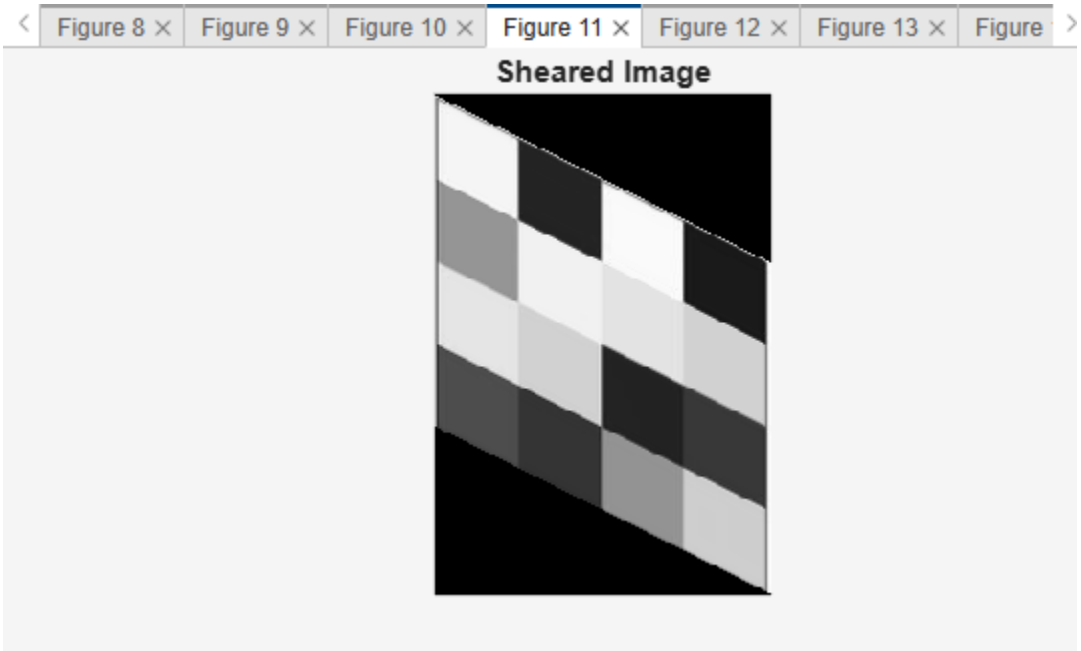


FIGURE 12:

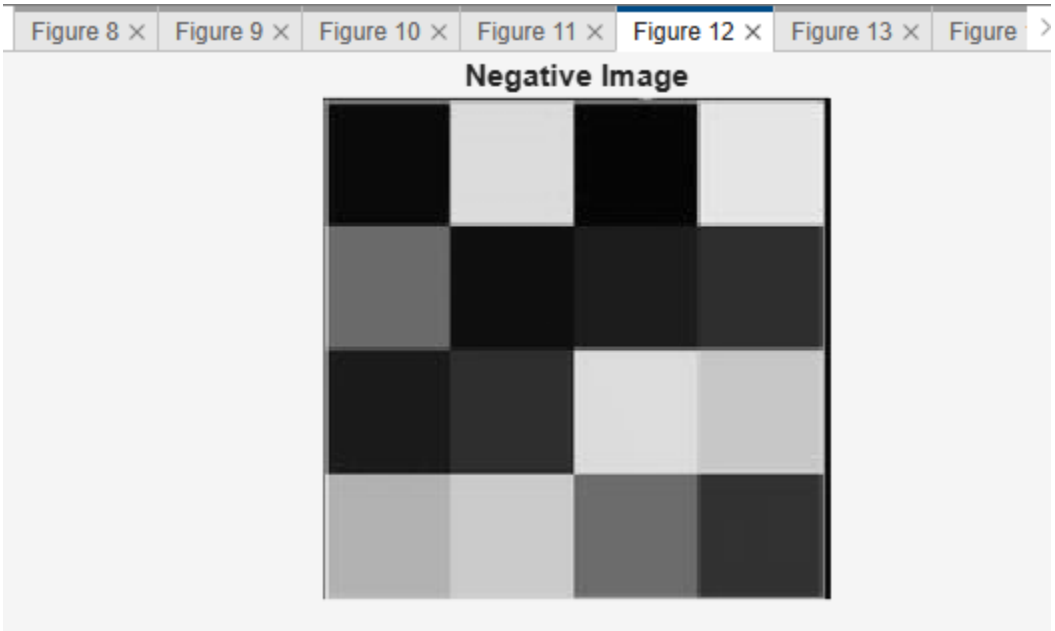


FIGURE 13:

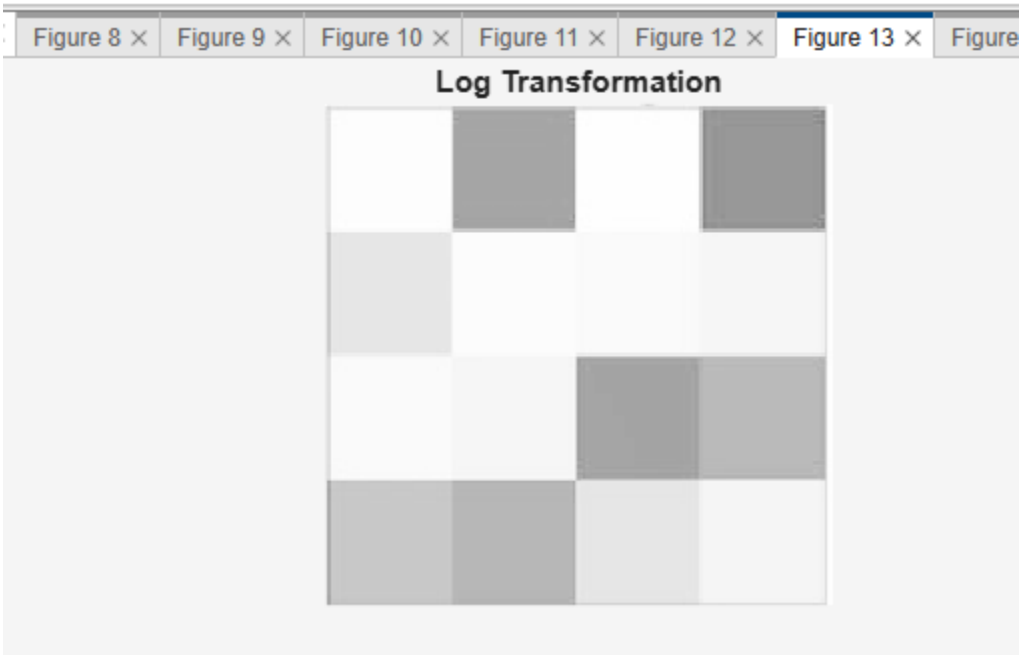


FIGURE 14:

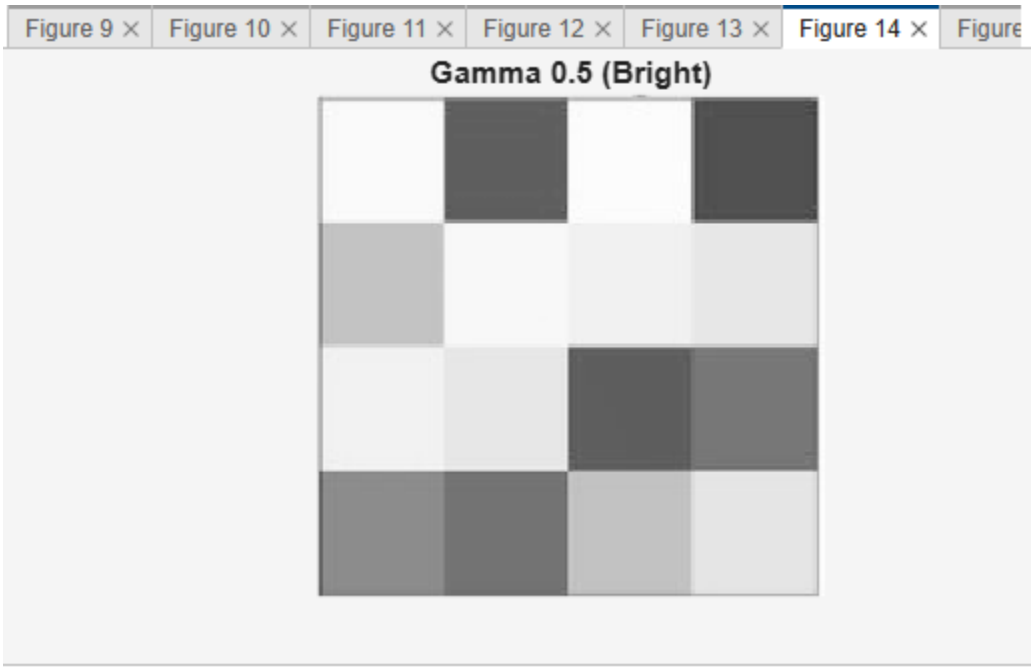


FIGURE 15:

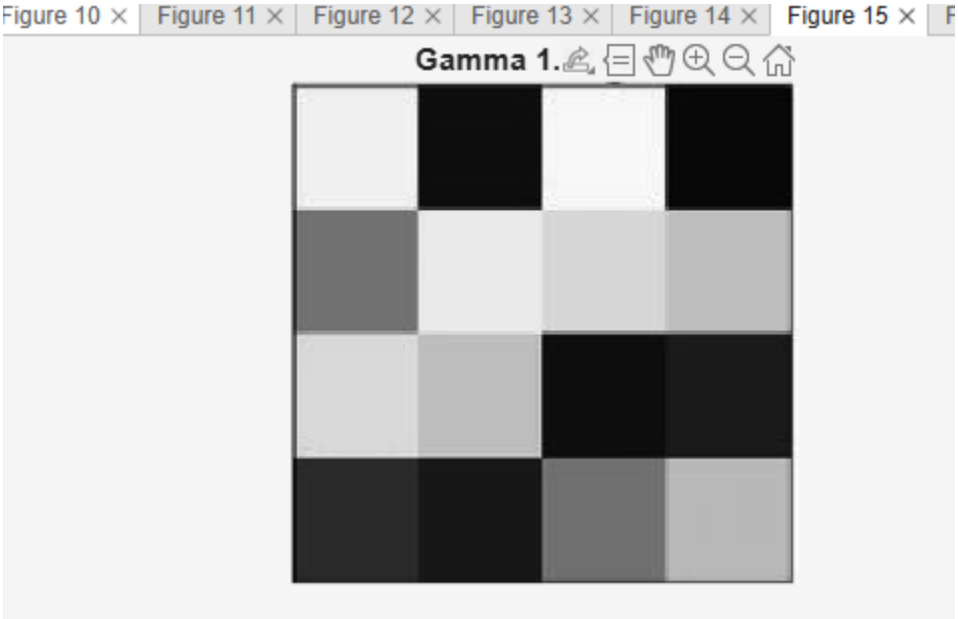


FIGURE 16:

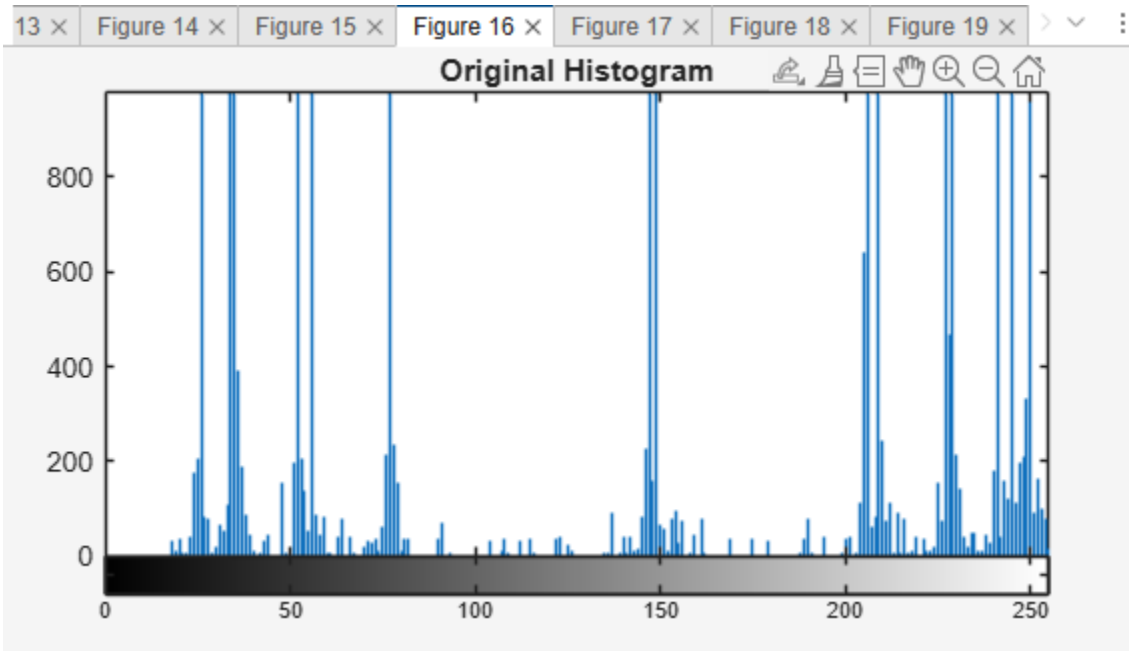


FIGURE 17:

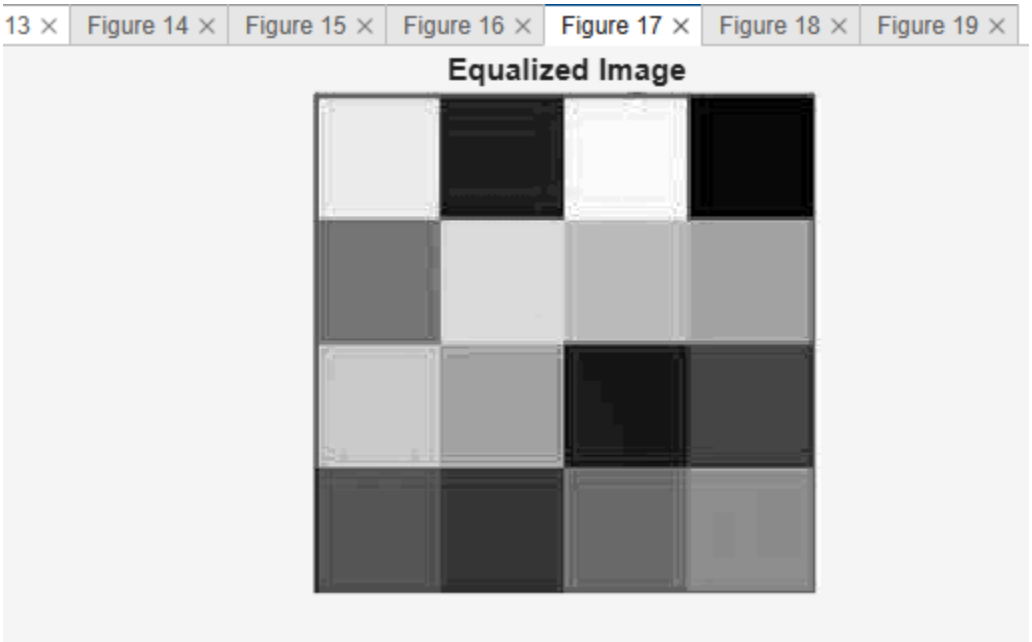


FIGURE 18:

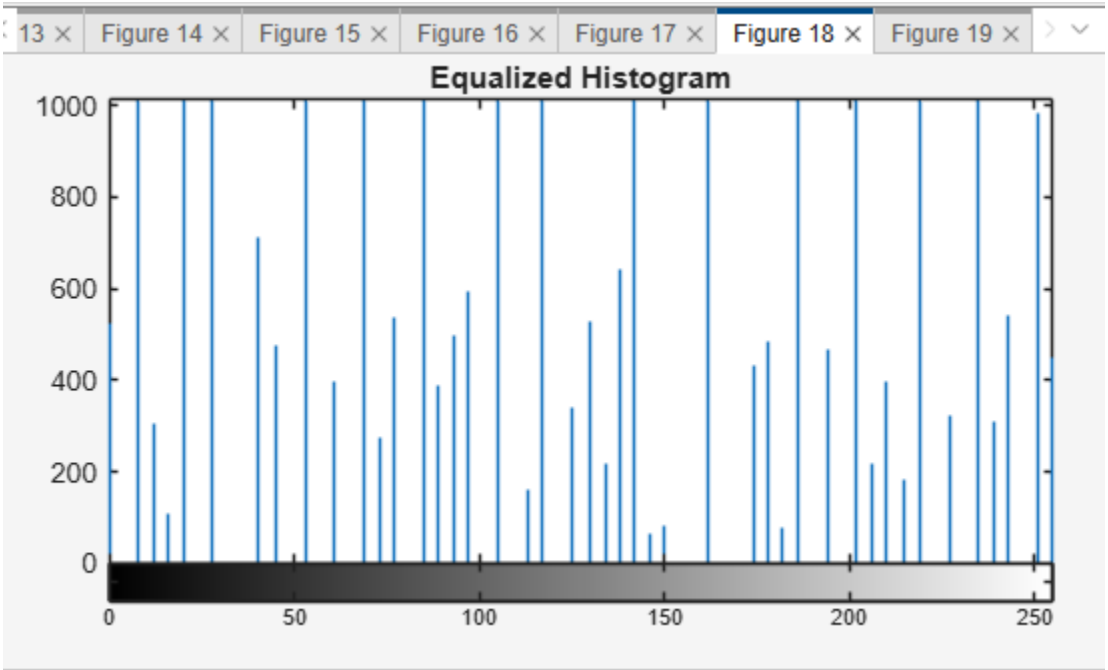
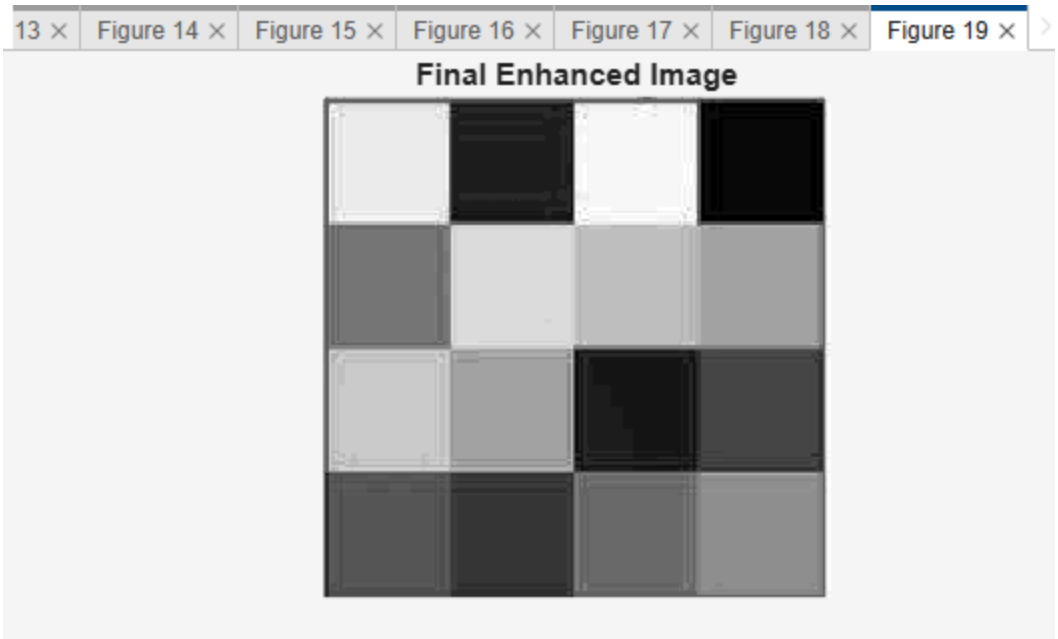


FIGURE 19:



QUESTIONS & ANSWERS

1. Why does histogram equalization improve contrast?

It spreads pixel intensities across the full range, making dark and bright regions more distinguishable.

2. How does gamma affect brightness?

- $\gamma < 1 \rightarrow$ image becomes brighter
- $\gamma > 1 \rightarrow$ image becomes darker

3. What is the effect of quantization on image quality?

It reduces number of intensity levels, causing loss of detail and banding.

4. Which transformation is reversible and why?

Geometric transformations like rotation and translation are reversible if parameters are known**.**

5. How do transformations affect spatial structure?

They change pixel positions, altering image geometry while sometimes preserving intensity**.**

GITHUB LINK:

https://github.com/Fatima235118/DIP-Image-Enhancement-System

LINKEDIN LINK:

http://linkedin.com/feed/